

PROJECT REPORT

Lynx-30: Simulation-Driven Low-Power Long-Range Camera Trap System

1. Abstract

This project presents the design and development of a low-power, long-range wireless camera trap system based on ESP32-CAM. The system is optimized for motion-triggered image capture using a PIR sensor and infrared illumination for night operation. A high-efficiency power architecture based on the TPS63060 DC-DC converter ensures stable operation from a battery source. The entire system is designed using KiCad with a modular hierarchical schematic approach and is prepared for 2-layer and 4-layer PCB implementation. The design emphasizes low power consumption, modularity, and manufacturability.

2. Project Objectives

The main objectives of this project are:

- Design a low-power embedded vision system
- Implement long-range motion detection using PIR sensor
- Enable night vision using IR LED illumination
- Develop a stable power system from battery input
- Design full schematic and PCB using KiCad
- Prepare a manufacturing-ready hardware design
- Ensure modular design using parent/child schematic sheets

3. System Overview

The system is divided into four main subsystems:



3.1 System Architecture Description

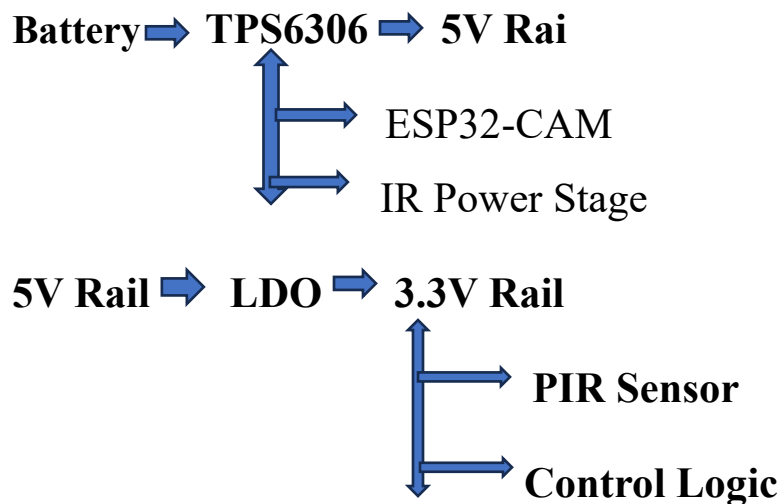
- The PIR sensor detects motion and sends a digital trigger signal.
- ESP32-CAM processes the signal and captures images.
- IR LEDs provide illumination during low-light conditions.
- Power system ensures stable voltage rails (5V and 3.3V).

4. Power System Design (CORE SECTION)

4.1 Power Architecture

The power system is built around a high-efficiency step-up/step-down regulator:

- Battery input feeds the system
- TPS63060 generates a stable 5V rail



4.2 Power Gating (IR Control)

IR illumination is controlled using a PMOS high-side switching circuit driven by an ESP32 GPIO signal.

- MCU_IR_EN controls PMOS gate
- IR LEDs are powered from 5V rail
- Ensures low power consumption during idle state

4.3 Design Considerations

- High efficiency switching regulator selected (TPS63060)
- Separation of analog and digital power domains
- Use of decoupling capacitors near ICs
- Ground plane used to reduce EMI and noise

5. Control System (ESP32-CAM)

5.1 Microcontroller Selection

The system uses **AI Thinker ESP32-CAM module**, which integrates:

- Camera sensor
- WiFi connectivity
- MicroSD interface
- Embedded ESP32 processor

5.2 Key GPIO Assignments

Function	GPIO
PIR Input	GPIO16
IR Control	GPIO4

Function GPIO

SD Card Clock GPIO14

SD Card CMD GPIO15

SD Data GPIO2, 4, 12, 13

5.3 Design Notes

- GPIO0 used carefully for boot mode control
- UART programming header included
- Pull-up resistors used for stable boot behavior

6. Sensor System (PIR)

A PIR motion sensor is used for detecting movement.

- Operating voltage: 3.3V
- Output: Digital HIGH on motion detection
- Connected to ESP32 GPIO16
- Includes pull-down resistor for signal stability

Motion → PIR Sensor → PIR_SIGNAL → ESP32 GPIO16

7. IR Illumination System

Infrared LEDs are used for night vision support.

- Powered from 5V rail
- Controlled via PMOS transistor
- Driven by MCU_IR_EN signal
- Designed for low standby power consumption

8. PCB DESIGN (KiCad Implementation)

8.1 Design Methodology

The design was implemented using:

- Hierarchical schematic design (Parent/Child sheets)
- Modular subsystem separation
- Clean power and signal domain separation

8.2 PCB Stackup Options

2-Layer Design (initial version)

- Top: signals + components
- Bottom: ground plane

4-Layer Design (final version)

Layer 1 → Signal routing

Layer 2 → Ground plane

Layer 3 → Power plane

Layer 4 → Signal routing

8.3 Layout Strategy

- TPS63060 placed close to input capacitors
- Ground plane kept continuous
- IR power traces isolated
- ESP32 antenna keep-out zone respected
- Short routing for switching regulator loop

8.4 Via Usage

- Vias used for GND stitching

- Power vias used for 5V and 3.3V distribution
- Signal vias used for layer transitions

9. Manufacturing Preparation

The design is prepared for fabrication using standard PCB manufacturers.

Deliverables include:

- Gerber files (all layers)
- Drill files
- BOM (Bill of Materials)
- Pick-and-place file
- PCB 3D render images

Manufacturing compatible with:

- JLCPCB
- PCBWay
- Seeed Studio

10. Simulation & Design Tools

The project was developed using:

- KiCad

11. Project Deliverables

- Complete schematic design (hierarchical)
- PCB layout (2-layer and 4-layer ready)
- Power architecture design
- IR illumination control system
- PIR motion detection system

- Manufacturing-ready Gerbers
- BOM and assembly files
- Project documentation

12. Conclusion

The Lynx-30 camera trap system demonstrates a complete embedded hardware design workflow from system architecture to PCB manufacturing readiness. The project integrates power electronics, embedded systems, and PCB design principles into a unified low-power IoT vision system. The modular design approach ensures scalability and future enhancement potential such as AI-based detection or cloud connectivity.